2026 Andes Virus MV Hondius Outbreak

Bayesian reconstruction of transmission chains using outbreaker2

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Outbreak summary · 11 cases in analysis · 3 deaths · 03 Apr – 14 May 2026

Background

Andes virus (ANDV) is the only hantavirus documented to transmit person-to-person, in addition to the usual rodent-to-human route (Alonso et al, 2020). The 2026 outbreak is linked to the **MV Hondius** expedition cruise ship. Cases first appeared in early April 2026. By the time whole-genome sequencing confirmed ANDV as the causative agent, passengers and crew had dispersed across Europe, Asia, and the Americas.

Methods

Data. 11 *confirmed* and *probable* cases were included in the analysis. WHO case IDs `12` and `13` were excluded as their symptom onset dates were unavailable. L-segment sequences were retrieved from Pathoplexus for 4 of the 11 cases with deposited sequence accessions matching the linelist's `accession_id`.

Cases with a recorded `ship_board_date` contributed a timed-contact likelihood to the model, representing the ship as a shared exposure setting from boarding until the earliest date of disembarkation, isolation or death.

Model. Transmission trees were inferred using `outbreaker2`, a Bayesian framework for reconstructing transmission chains from epidemiological, contact, and genetic data (Jombart et al, 2014; Campbell et al, 2018). The serial interval was modelled as a discretised Gamma distribution with mean 18 days and coefficient of variation (CV) = 1/3 (UKHSA rapid report, 2026). Up to two unobserved intermediate generations were allowed between observed cases (`max_kappa = 3`).

The reporting rate π was assigned a Beta(10, 1) prior with an initial value of 0.95, reflecting the expectation of high case ascertainment in a closed cruise cohort. Contact reporting coverage (ε) and place persistence (τ) were fixed at 1, assuming that recorded ship contacts were complete and that the ship was the only shared transmission setting during the voyage.

Because the L-segment alignment showed near-zero genetic variation (see SNP matrix in Results), the per-site mutation rate μ could not be reliably estimated from the data and was therefore fixed at 1.7×10^{-5} substitutions/site/generation. This value was derived from the published ANDV substitution rate of $\sim 3.5 \times 10^{-4}$ substitutions/site/year (Ramsden et al, 2008) and an 18-day serial interval.

Inference. Four independent MCMC chains were run for 100,000 iterations each, with samples retained every 50 iterations. The first 10,000 iterations of each chain were discarded as burn-in. Convergence was assessed using three complementary diagnostics: the Gelman–Rubin $\hat{R}$ statistic and effective sample size for scalar parameters, and the PERMANOVA test implemented in mixtree `mixtree` (Geismar et al, 2025). Consensus ancestries were defined as the modal inferred infector for each case across the pooled posterior distribution. Ancestry uncertainty was summarised using Shannon entropy (o2ools package).

Results

Timeline of Exposure Windows

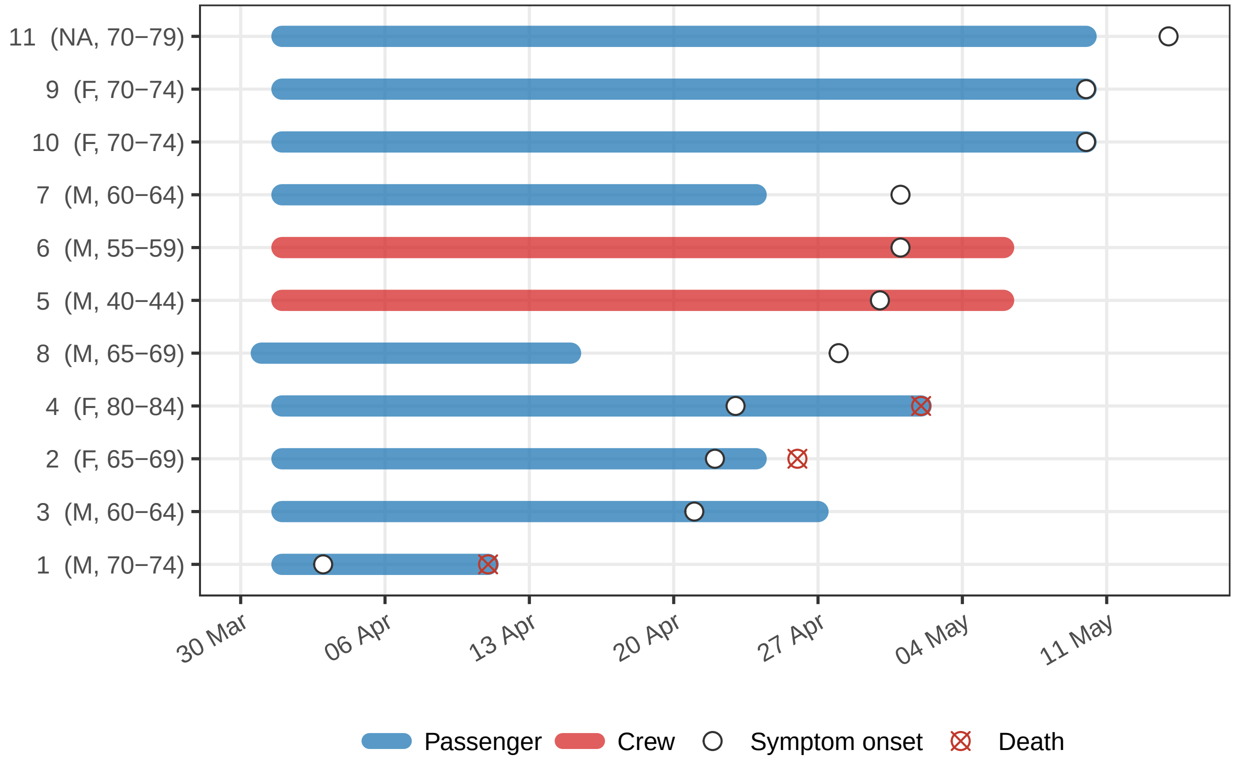

Figure 1: Exposure windows per case defined by time spent aboard the MV Hondius. Segments represent periods of presence on the ship, with colours indicating role. Circles indicate dates of symptom onset, and crosses indicate dates of death.

Genomic Analysis

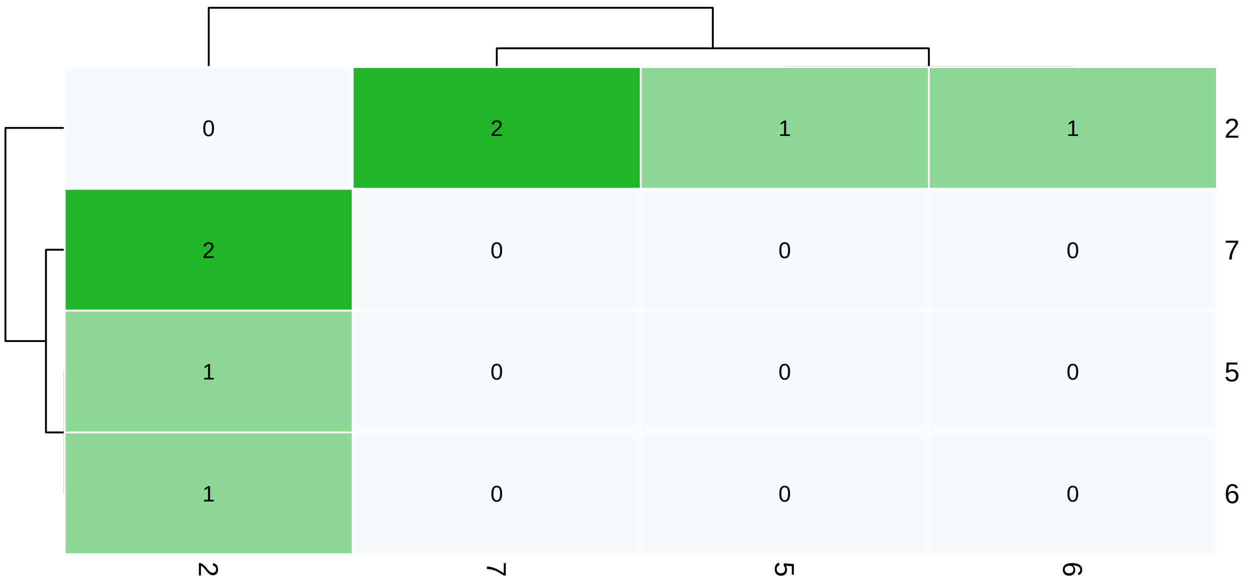

Figure 2: Pairwise SNP distance matrix for sequenced cases (L segment). Values are raw nucleotide differences.

Model Diagnostics

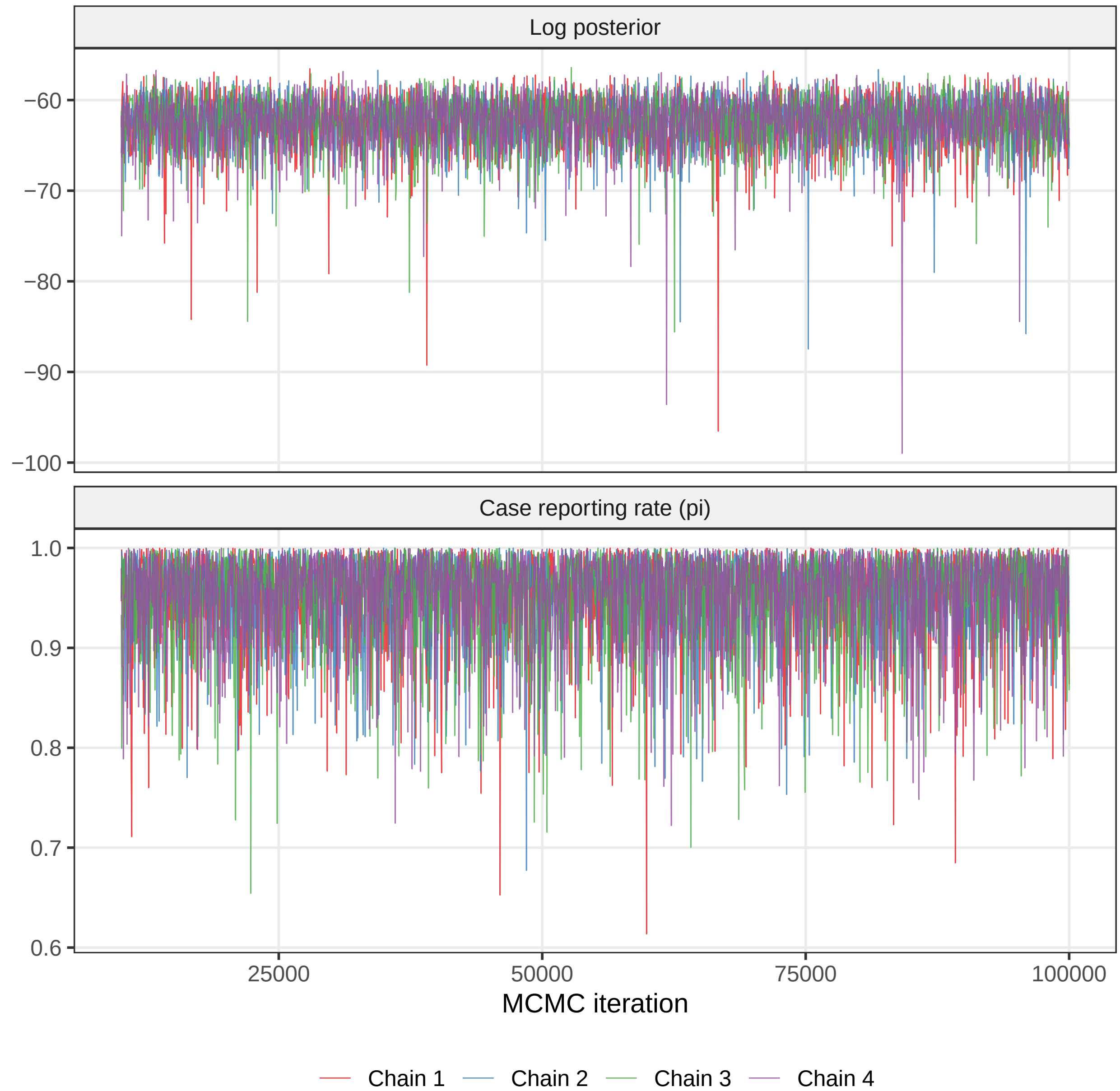

Figure 3: MCMC trace plots for scalar parameters (post-burnin). Stationary, well-mixed traces indicate convergence.

Table 1: Convergence diagnostics across 4 independent chains.

Parameter	R-hat	ESS
Case reporting rate (pi)	1.002	7,200
Infection times (t_inf, N = 11)	1.001	5,242

ESS: effective sample size out of 7,200 total posterior samples (4 chains of 1,800 samples each). The ***t_inf*** *row shows the worst-case (max R-hat, min ESS) across all per-case infection times.*

Table 2: PERMANOVA test results.

df	F	R^2	p-value	Permutations
3	0.97	0.004	0.531	999

The topology of transmission trees, defined by the arrangement of vertices and edges, encodes key epidemiological characteristics. PERMANOVA compares the distributions of trees between forests (i.e. sets of trees) using pairwise tree distances (here, the patristic distance). A non-significant result ($p > 0.05$) indicates that there is no evidence of a difference in the topological distributions of the forests under the chosen distance metric (Geismar et al, 2025). In the context of MCMC diagnostics, this provides evidence that independent chains have converged to the same posterior distribution of transmission trees.

Transmission Tree

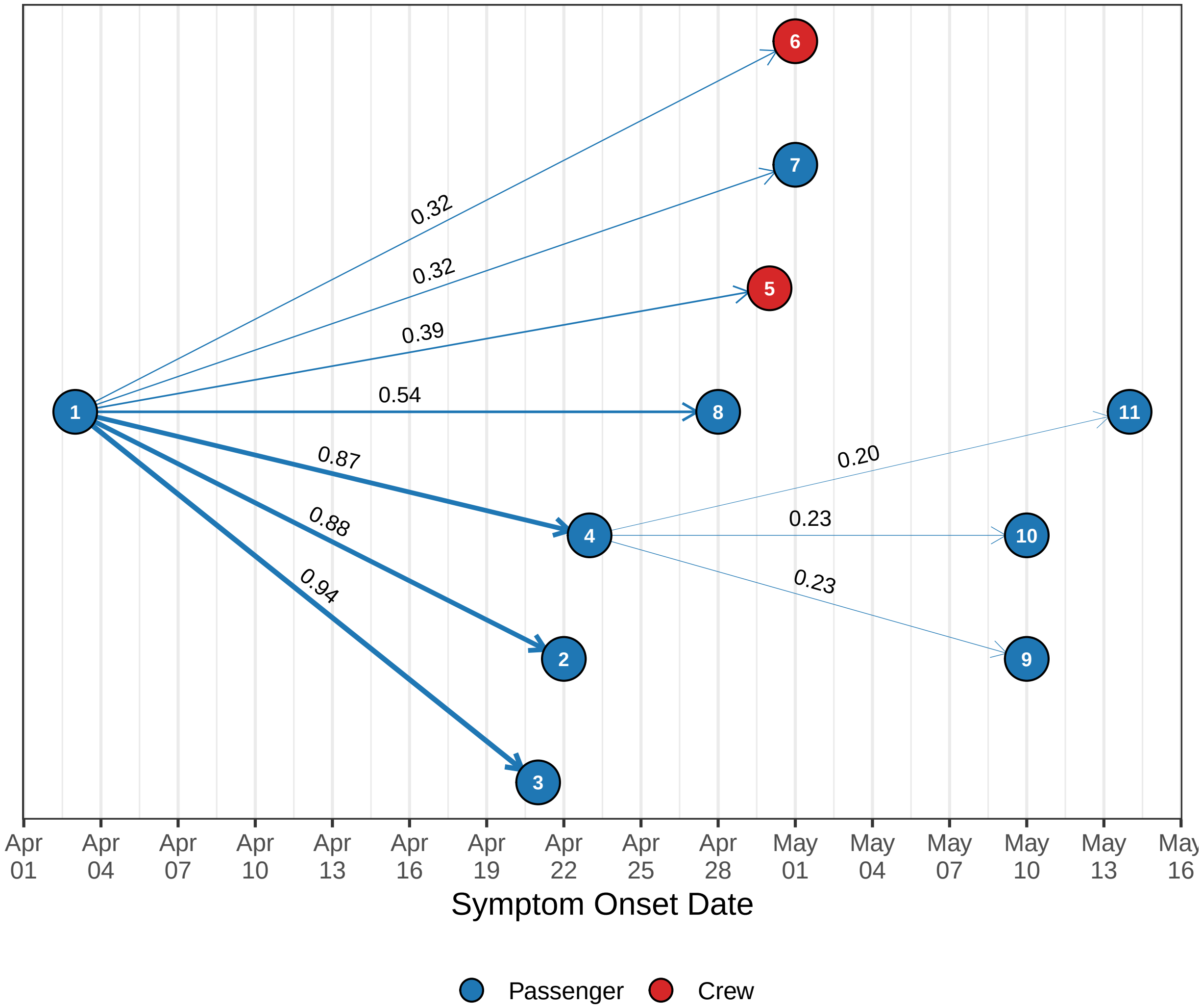

Figure 4: Consensus transmission tree, where each case is assigned its most frequently inferred ancestor across the posterior sample.

Ancestry Uncertainty

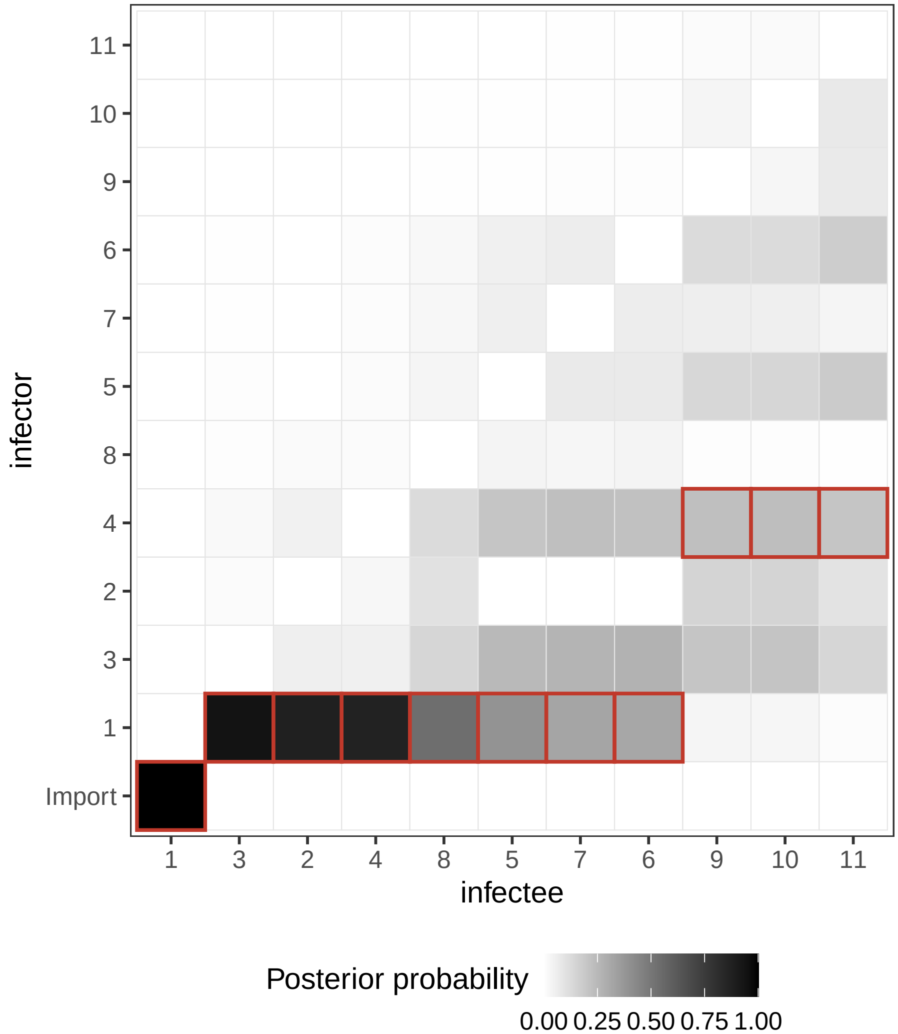

Figure 5: Ancestry support matrix: for each case (x-axis), the tile shading shows how often each potential infector (y-axis) was sampled across the posterior. Red outlines mark the consensus-tree ancestor (modal infector). 'Import' aggregates samples where the case was assigned no ancestor.

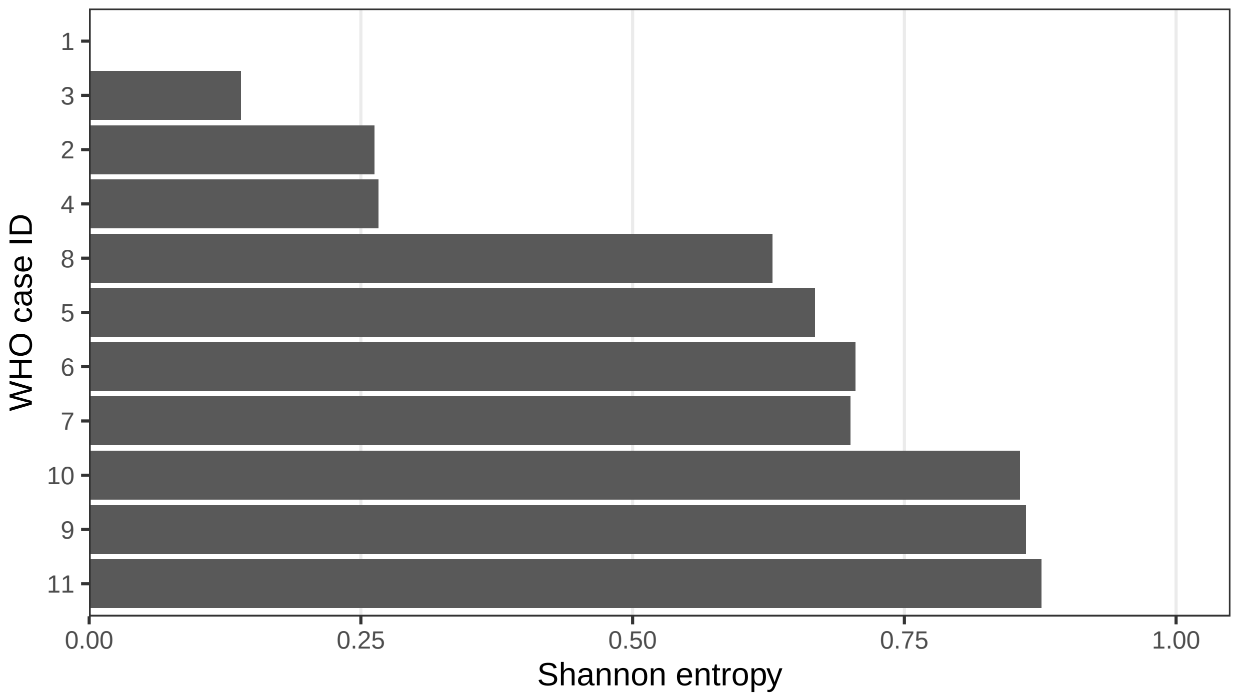

Figure 6: Shannon entropy of each case's inferred source. Entropy = 0 means a single unambiguous infector. See o2ools::get_entropy() for details.

Dates of Infection

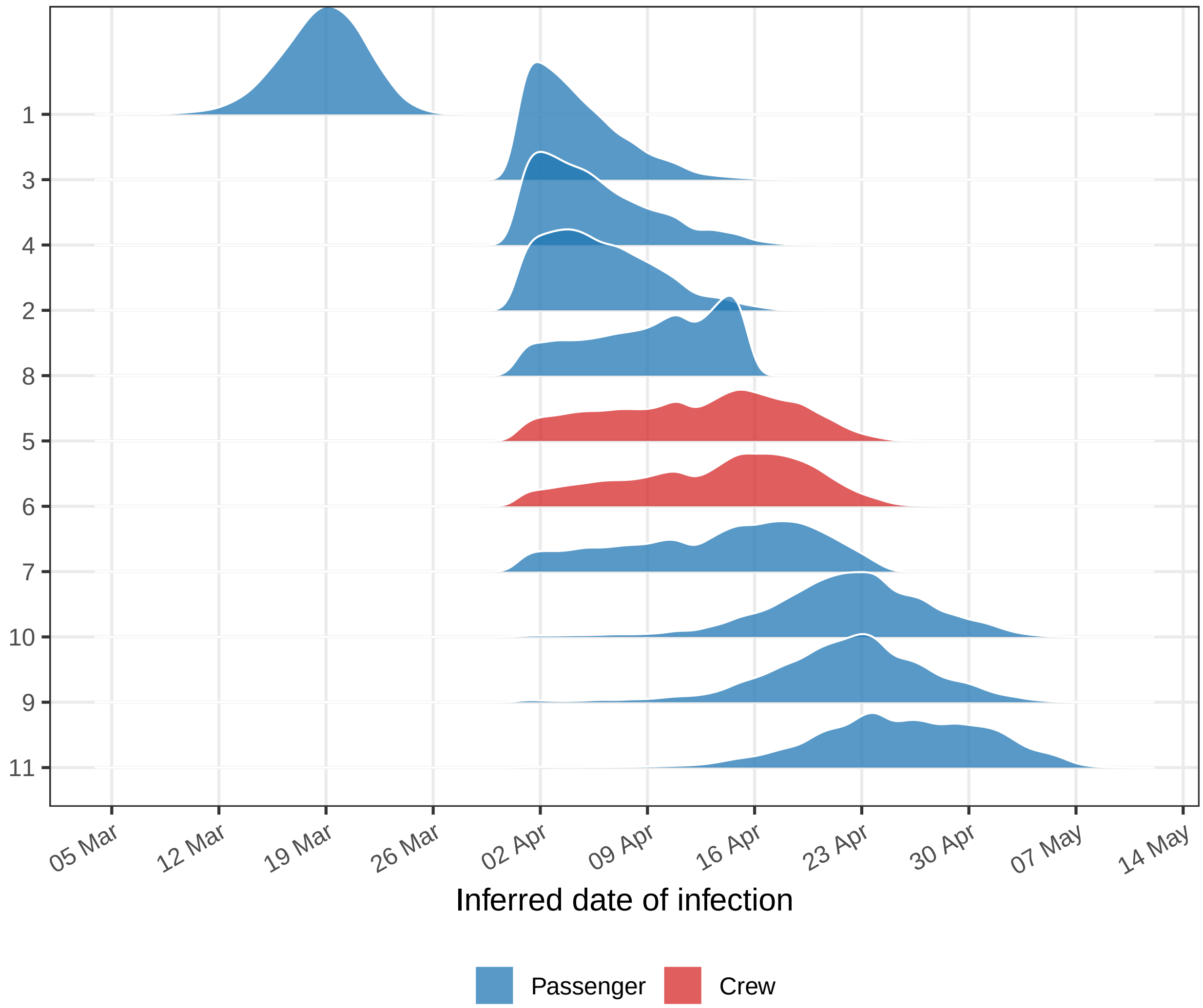

Figure 7: Posterior distribution of inferred infection dates per case. Width reflects uncertainty; cases ordered by mean inferred infection date.

Imports and Unobserved Transmission

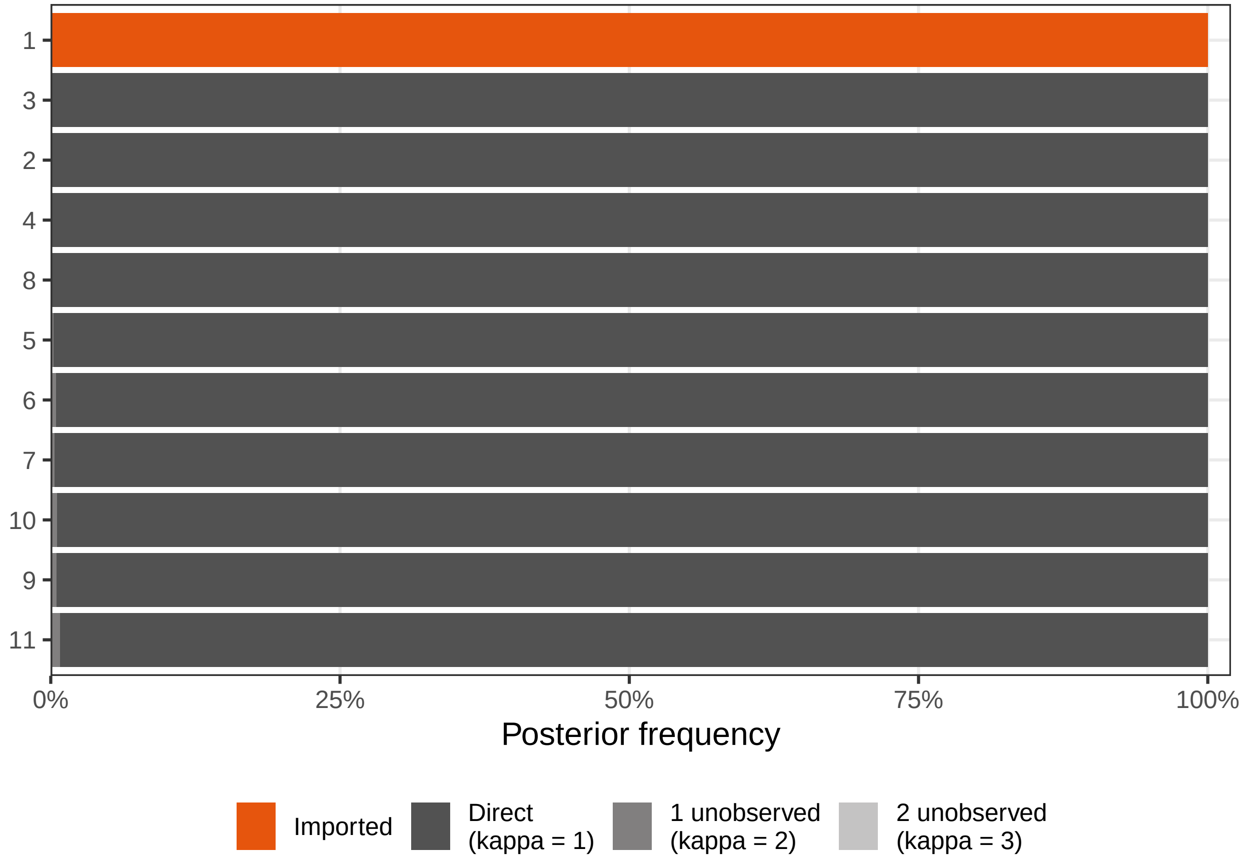

Figure 8: Posterior probability of each case being imported, directly linked, or connected via unobserved intermediate cases.

Offspring Distribution

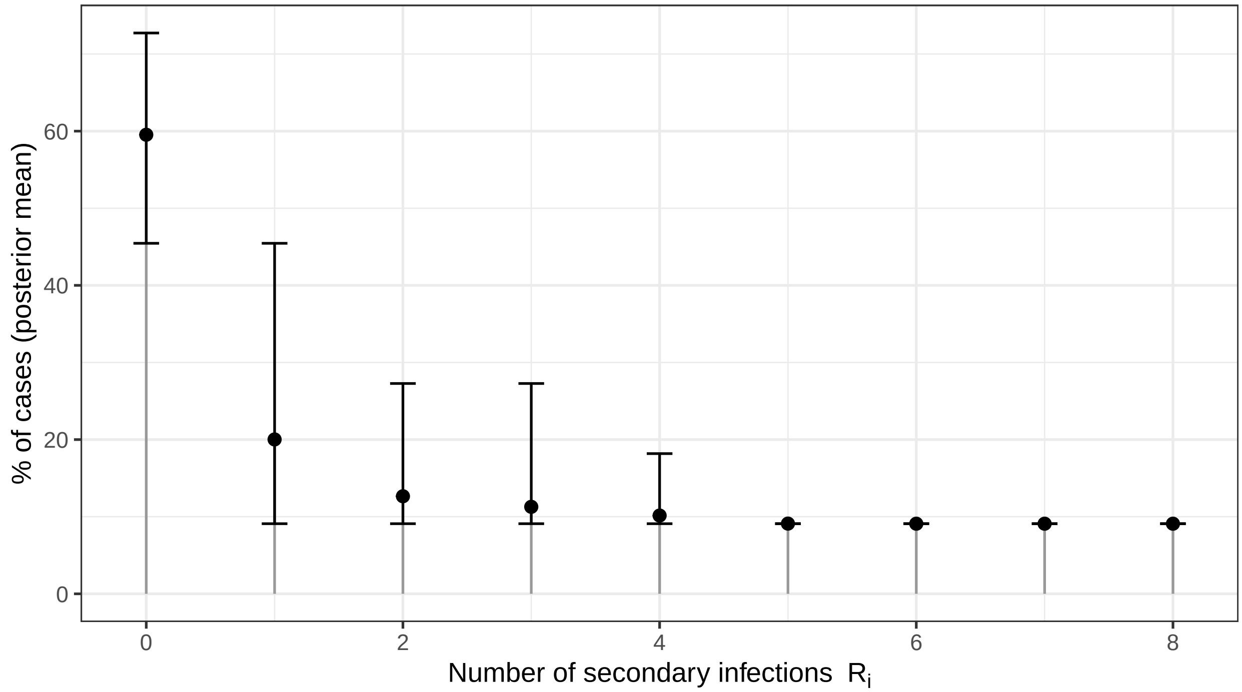

Figure 9: Posterior distribution of the offspring distribution, representing the distribution of secondary infections caused by an infected individual.

Discussion

The reconstruction supports a single introduction of ANDV onto the MV Hondius, with case `1` as the likely index case and source of an early superspreading event. Three secondary infections attributed to case `1` (cases `2`, `3` and `4`) are strongly supported (posterior probability ≥ 0.85), while four additional secondary infections (cases 5, 6, 7 and 8) are plausible but more uncertain (posterior probability 0.32–0.54). A subsequent generation of transmission, potentially seeded by case `4`, received considerably weaker support (posterior probability 0.20–0.23 for transmission to cases `9`, `10` and `11`) and should therefore be interpreted cautiously.

Two factors limited reconstruction resolution. First, cases `12` and `13` were excluded because symptom onset dates were unavailable and only confirmation dates had been recorded. Second, the L-segment alignment exhibited almost no genetic variation, meaning that genomic data contributed little information beyond confirming that sequenced cases were epidemiologically linked. Together, these limitations explain the substantial uncertainty surrounding inferred infectors beyond the initial cluster.

Despite these limitations, the reconstruction is most consistent with a single introduction followed by an early superspreading event and, at most, one additional generation of transmission potentially seeded by case `4`. There was little evidence for transmission through unobserved intermediate cases.

Updated line list data, including symptom onset dates for cases `12` and `13`, or alternatively their sampling dates, would be valuable. Additional sequence data may also help resolve the remaining ambiguous transmission chains.

Data

Data available under CC BY 4.0 at kraemer-lab/Hondius_hantavirus_h2026. Sequences retrieved from Pathoplexus. Report source: `report.qmd` available at CyGei/hantavirus_26.